

Arsat.P
192424038

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Arsat.P/192424038 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Arsat.P

level -4
60

Annexure A

Student 1: AKILAN SP | Register Number: 192212001

1. Solve the Missionaries and Cannibals problem: 3 missionaries and 3 cannibals must cross a river using a boat that can carry at most 2 people, such that cannibals never outnumber missionaries on either bank. Formulate the problem as a search problem (state, initial state, actions, transition model, goal test) and show the first three states explored using Breadth-First Search.
2. An e-commerce company wants to deploy a delivery drone to autonomously deliver packages within a city. Identify what type of problem-solving agent is suitable for this scenario, and write pseudocode outlining how the agent formulates and solves the delivery routing problem.
3. Given a graph with 8 nodes and a start node S and goal node G, trace Depth-Limited Search (DLS) with a depth limit of 3. Show which nodes are expanded and explain what happens if the goal lies beyond the depth limit.

Student 2: AKULA GANESH RAGHAVENDHRA | Register Number: 192312116

1. Consider the map of Romania (Arad to Bucharest) as a graph of cities connected by roads with distances. Trace the order in which cities are expanded using Breadth-First Search and then Depth-First Search, starting from Arad. Compare the two traces in terms of completeness and optimality.
2. For a Tic-Tac-Toe playing agent, define its PEAS description and characterize the environment (is it fully observable? deterministic? episodic? single or multi-agent?). Explain why this environment is classified as adversarial.
3. A smart thermostat adjusts room temperature based on current readings and a target temperature. Design it first as a Simple Reflex Agent, then redesign it as a Model-Based Reflex Agent. Explain what internal state the model-based version must maintain and why it performs better.

Student 3: ARSAT P | Register Number: 192424038

1. Solve the Water Jug problem with a 5-gallon jug and a 3-gallon jug (no measuring markers). How can you measure exactly 4 gallons of water using only these two jugs and a pump? List your explicit assumptions and show the sequence of states from start to goal.
2. Given a graph with 8 nodes and a start node S and goal node G, trace Depth-Limited Search (DLS) with a depth limit of 3. Show which nodes are expanded and explain what happens if the goal lies beyond the depth limit.
3. Trace the working of Breadth-First Search and Uniform Cost Search on a weighted graph with at least 7 nodes where edge costs are unequal. Show why BFS may not find the optimal (least-cost) path while UCS does, using your example graph.

2. Formulate the 8-puzzle problem (a 3x3 sliding tile puzzle) as a search problem by specifying the state representation, initial state, goal test, and the set of possible actions. Explain why the branching factor of this problem is at most 4.
3. For a Tic-Tac-Toe playing agent, define its PEAS description and characterize the environment (is it fully observable? deterministic? episodic? single or multi-agent?). Explain why this environment is classified as adversarial.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Assessment

COI-ATI

NAME:- Arsalan P

Reg no:- 192424038

Course:- CSA1714

1) Water Jug problem (5-Gallon and 3-Gallon Jug)

Problem:

using a 5-gallon jug and a 3-gallon jug (without measuring mark), measure exactly 4 gallon of water

Assumption:

- * Both jugs are Initially empty
- * There is an unlimited water supply
- * Water can be poured from one jug to another until one jug is either full or empty
- * Water can be discarded whenever needed

Sequence of state:

Step	Action	State (5G, 3G)
Start	Both empty	(0, 0)
1	Fill 5G jug	(5, 0)
2	Pour 5G into 3G	(2, 3)
3	Empty 3G	(2, 0)
4	Pour remaining 2G into 3G	(0, 2)
5	Fill 5G again	(5, 2)
6	Pour from 5G to 3G is full	(4, 3)

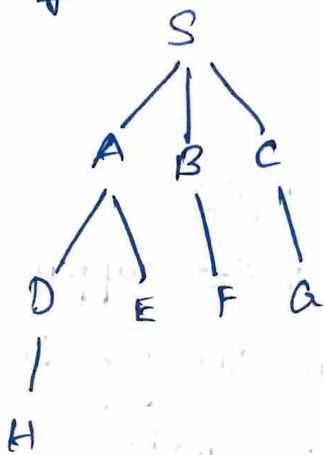
Goal State

$\therefore (4, 3)$

$\therefore$ The 5-gallon jug now contains exactly 4 gallons, which is the required solution.

2) Depth - Limited Search (DLS)

Given graph:



Depth limit = 3

Node - Depth:

Node	Depth
S	0
A, B, C	1
D, E, F, G	2
H	3

Suppose the goal node is G.

Traversal order:

- (i) Expand S
- (ii) Expand A
- (iii) Expand D
- (iv) Expand H
- (v) Backtrack
- (vi) Expand E
- (vii) Backtrack
- (viii) Expand B
- (ix) Expand F
- (x) Backtrack
- (xi) Expand C
- (xii) Expand G (Goal Found)

Expanded Node:

$S \rightarrow A \rightarrow D \rightarrow H \rightarrow E \rightarrow B \rightarrow F \rightarrow C \rightarrow G$

Case:- Goal Beyond Depth Limit

Suppose another node I exists below H

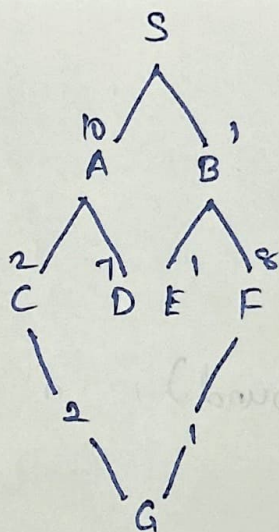
S
|
A
|
D
|
H
|
I (Depth 4)

Conclusion

- * DLS searches only up to the specified
- * It saves memory compared to BFS.
- * If the Goal is deeper than the limit, DLS report failure (Cutoff), even when a valid Solution exist

3) Breadth - First Search (BFS) vs uniform cost search

Weighted Graph:



Edge Costs:

Edge

Costs

S → A

10

S → B

1

A → C

2

A → D

7

B → E

1

B → F

8

C → G

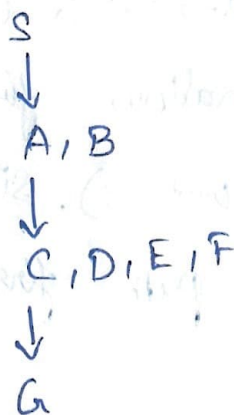
2

F → G

1

Breadth-First Search (BFS)

Traversal:



Suppose BFS searches:

$S \rightarrow A \rightarrow C \rightarrow G$

No. of edge = 3

Total Cost = $10 + 2 + 2$
 $\Rightarrow 14$

Uniform Cost Search (UCS)

UCS always expand the node with the lowest cumulative Path cost

Expansion:

Step	Expanded Node	Cost
	S	0
1	B	1
2	E	2
3	A	10
4	C	12
5	G	14

In this graph, UCS also reaches $S \rightarrow A \rightarrow C \rightarrow G$ with cost 14 because no cheaper path exists.

* BFS may still return $S \rightarrow A \rightarrow C \rightarrow G$ (cost = 14) because it encounters that shallow first

* UCS returns $S \rightarrow B \rightarrow E \rightarrow G$ (cost = 3). Since it always expands the lowest-cost path first

Comparison:

Feature	BFS	UCS
Expansion	Level by level	Lowest Cumulative Cost first
Edge Cost	Ignores weights	Uses edge weights
Optimal	Only when all edge cost are equal	Yes, for positive edge cost
Data structure	Queue (FIFO)	Priority Queue

Conclusion:

* BFS:- Finds the solution with the fewest number of edges but may not produce the least-cost path edge

Cost differ

* UCS:- always returns the minimum-cost path because it expands node based on the accumulated path cost rather than depth